

DOCUMENT 01

Complete decision register

Confirmed requirements, working directions, deferred ideas and open choices

1. Status and scope

This is the complete discussion register for the three user requests in this conversation. It is not a claim that the app has been changed. Research date: 23 September 2026. Sources were inspected on public main/master branches; an immutable Manga Cleaner commit was not established.

Status key: CONFIRMED means explicitly requested by the user. WORKING DIRECTION means an assistant recommendation consistent with the request, but not a benchmarked or individually approved choice. DEFERRED preserves an earlier idea without making it a dependency. OPEN requires a product choice, experiment or permission check.

Confirmed requirements

C01 - Clean the text properly. The goal is removal of lettering, not transcription. Regular bubble text and sound effects are both in scope.

C02 - Recover missed regions. The user reports that most SFX and some bubble text are missed. Investigate true detector misses separately from detected-but-skipped regions and inadequate erase masks.

C03 - No OCR requirement. This workflow must not require manga-ocr, text recognition, translation, a transcript, or a successful language read. Existing optional reading features need not be deleted from the app.

C04 - Text-shaped affected area. Add a selection/cleaning area that follows the detected lettering or compact text footprint instead of repainting a large surrounding rectangle.

C05 - Small adjustable padding. Expand that footprint slightly around the text. The user has not selected a numeric default, unit convention or maximum.

C06 - The visible footprint governs the edit. The new highlighted area is the area within which changes may be applied. It must not merely decorate a larger hidden erase rectangle.

C07 - Preserve the old option. The new behavior is additional. Do not replace, silently migrate or remove the current cleaning/selection behavior.

C08 - Use the images as references. The first image communicates the desired compact shape; the second communicates unwanted rectangular overreach. They are illustrative references, not a verified before/after app output or pixel-accurate training labels.

C09 - Deliver itemized documentation. Preserve the full discussion and research, including earlier alternatives, caveats and decisions still open, in separate numbered documents.

Current scope interpretation

The earlier request for class detection for “EVERYTHING” is now an operational goal: find and clean all target lettering safely. A large semantic taxonomy is not required to deliver this feature. Detection and pixel segmentation are still required; “no OCR” does not exclude those image-processing tasks.

2. Working technical directions

The following are proposed implementation decisions. They are not presented as already approved changes or measured improvements.

ID	Working direction
R01	Retain CTD as the current text discovery and segmentation baseline.
R02	Evaluate a COO-trained detector for extra SFX coverage; start with DBNet++ detection-only. CTD + COO is the leading experiment, not a proven winner.
R03	Keep the existing ogkalu text/bubble detector for enclosure and complementary candidates; it is neither OCR nor a detailed SFX classifier.
R04	Add a no-recognition, all-text cleaning policy. Bypass the Japanese gate only for that explicitly selected policy; do not remove legacy gating globally.
R05	Keep detection profile, cleaning-area mode, language policy and outside-bubble permission independent.
R06	Retain credible candidates from either detector. Do not require CTD agreement to admit a COO-only SFX.
R07	Test CTD's unused line output, orphan segmentation candidates and overlapping high-resolution passes for missing bubble text.
R08	Refine lettering at source resolution using CTD evidence and the source image; evaluate upstream-style refinement before adding another large model.
R09	Store an immutable base lettering mask. Derive padding from that base, never from the previously padded/applied mask.
R10	Separate detection bounds, refinement/inpainting context, the model's hole and final allowed write support.
R11	In the new mode, every participating engine must respect one authoritative final mask; do not add invisible isolation or denoise margins afterward.
R12	Show the same backend-generated footprint in the UI, use it for cleaning, and persist its identity/hash. Rectangles remain metadata and optional handles.
R13	Preserve disconnected SFX fragments, punctuation and relevant outlines/shadows. Grouping does not authorize filling the space between fragments.
R14	Provide mask correction and explicit review outcomes. An empty or unsafe mask must not silently become a filled detection rectangle.
R15	Preserve layers, undo/redo, source files, export fidelity, region provenance and old-project behavior.
R16	Prototype research checkpoints separately. Ship an optional native runtime adapter only after output parity, device testing and licensing review.
R17	Evaluate complementary recall, artwork damage and human cleanup effort, not only OCR accuracy or detector leaderboard Hmean.
R18	Implement strict mask geometry first; additional SFX detection must feed that safe path rather than make larger erasures.

3. Earlier ideas retained, not silently promoted

ID	Discussed item and current disposition
D01	Manga-ocr together with COO: technically a different recognition stage, but no recognition is needed for the user's current workflow. Keep old optional support; do not add it to the new dependency chain.
D02	PP-OCRv6 medium detector: optional additional discovery experiment, especially on non-Japanese or missed small text; its recognizer is out of scope.
D03	Other ogkalu checkpoints: an A/B test, not an assumption that a larger or non-quantized export wins. Validate each export's contracts.
D04	MTSv3 / ABCNetv2: research comparators, not default bundled choices. Deployment and licensing need separate checks.
D05	COO TRBA recognition and M4C linking: not needed for cleaning. Geometric grouping can exist without recognizing text.
D06	ogkalu YOLOv8m text segmenter: optional mask experiment; verify whether masks represent text strokes or broader text regions.
D07	SAM-TS / Hi-SAM: newly researched stroke-segmentation comparator, not an approved production dependency.
D08	Detailed semantic-role classifier: defer unless users need different treatment of dialogue, SFX, signage, credits or other roles.
D09	Magi v2/v3, MangaLMM and vision-language reviewers: preserve as earlier exploration; exclude from the immediate cleaning feature.
D10	Custom model training: future option if evaluation shows persistent misses or poor masks; requires usable rights and independently annotated holdouts.
D11	Full-page plus roughly 1024-1536 source-pixel crops with 15-25% overlap: earlier experimental starting range, not an agreed optimal setting.
D12	Text-edge padding around 1-3 source pixels: earlier experimental starting range, not a fixed product decision.
D13	Automatic residual-text checks: may use visual or detector evidence without OCR; uncertain results still need human review.
D14	Heavier inpainting/cloud engines: no new requirement. Preserve existing availability/consent rules; do not make them automatic dependencies.

Earlier taxonomy, preserved for future work

Role: dialogue, thought, narration, aside, SFX, scene text/signage, heading/title, editorial note, credits, page number, watermark/logo text, ruby/furigana, unknown text. Punctuation-only text remains eligible.

Independent attributes: enclosure (speech/thought bubble, caption, uncontained, uncertain); script/language evidence (Japanese, Chinese, Korean, Latin, mixed, unknown, kept separate where possible); appearance (printed, handwritten, outlined, reversed, vertical, rotated, curved); relations (inside container, furigana parent, effect fragment, continuation); action (clean, keep, review) and mask quality.

These labels were proposed, not learned capabilities of the current models. They are not a prerequisite for reliable cleaning.

4. Corrections, qualifications and remaining choices

Corrections to carry forward

- Q01 - “COO” is not a universal model.** It is a dataset/research collection with multiple checkpoints. Detection polygons are not automatically lettering-accurate erase masks. See Document 04 and sources S22-S24.
- Q02 - The example is Korean.** Its Hangul lettering makes multilingual validation necessary. Japanese COO results cannot establish Korean/color-webtoon recall. This qualifies the earlier CTD + COO recommendation; it does not reject the experiment.
- Q03 - The old path is not uniformly a filled rectangle.** Source inspection shows existing pixel masks and a separate narrower model-ink path. The user-facing rectangular overlay is not proof that every backend path fills that rectangle. Preserve legacy behavior rather than replacing it with a newly invented rectangle algorithm. See Document 02.
- Q04 - More padding alone is not the fix.** First make the base mask match lettering; then grow that mask. More padding on a coarse blob increases artwork exposure.
- Q05 - Mask quality and inpainting quality differ.** A correct mask limits where the app edits. It does not guarantee exact reconstruction of artwork already hidden by opaque lettering.
- Q06 - No performance claims were established here.** No checkpoint inference, desktop execution, code changes or app benchmark was performed in this research session. Published scores are not measurements on the user's references.

Open choice register

ID	Unresolved choice	Recommended starting position
O01	Public mode names and default	“Current area (legacy)” and “Text-shaped + padding”; keep current default until explicit selection.
O02	Glyph-tight versus compact grouped footprint	Start from glyphs. Preview small optional grouping/closing; never force a convex hull.
O03	Padding default, range and scaling	Test 0, 1, 2, 3 and 5 source pixels; consider 2 px only as a trial default.
O04	Outlines, shadows and glows	Remove decoration belonging to lettering; retain uncertain broad glows for correction.
O05	Target scripts and art styles	Include Japanese and Korean at minimum in evaluation; determine other production priorities.
O06	Which SFX checkpoint ships	Decide after incremental recall, false positives, device cost and permission review.
O07	Whether a second segmenter is needed	Compare refined CTD with SAM-TS/Hi-SAM and the comic segmenter on actual masks.
O08	Per-device latency/memory budgets	Measure on the app's intended platforms before promising budgets.
O09	Outside-bubble permission and resume UX	Explicit all-text preset, visible scope, and a clear resumption policy; no silent global policy change.
O10	Licensing / checkpoint redistribution	Complete the exact artifact review in Document 08 before bundling.

Package relationship: Documents 02-08 expand these requirements and proposals. Where a later implementation choice differs, update this register rather than treating a previous suggestion as user approval.

Source IDs link directly to primary references. Full evidence ledger: Document 08.